

## Qualification Pack



# AR/VR - Indiaskills Worldskills

QP Code: MES/Q0521

Version: 1.0

NSQF Level: 4.5

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## Qualification Pack

## Contents

|                                                                                                              |    |
|--------------------------------------------------------------------------------------------------------------|----|
| MES/Q0521: AR/VR - Indiaskills Worldskills .....                                                             | 3  |
| <i>Brief Job Description</i> .....                                                                           | 3  |
| Applicable National Occupational Standards (NOS) .....                                                       | 3  |
| <i>Compulsory NOS</i> .....                                                                                  | 3  |
| <i>Qualification Pack (QP) Parameters</i> .....                                                              | 3  |
| MES/N0562: Prepare model and complete the texture as per real-time engines requirement-<br>Worldskills ..... | 5  |
| MES/N0564: Test 3D models in the real-time/game environment- Worldskills .....                               | 8  |
| MES/N0565: Artificial intelligence & machine learning- Worldskills .....                                     | 11 |
| MES/N0567: Deploy Internet of things (IoT)- Worldskills .....                                                | 16 |
| MES/N0568: Enterprise block chain- Worldskills .....                                                         | 21 |
| MES/N0569: Prepare computer generated models- Worldskills .....                                              | 26 |
| MES/N0570: Add textures to models- Worldskills .....                                                         | 29 |
| MES/N0104: Maintain Workplace Health & Safety .....                                                          | 32 |
| DGT/VSQ/N0102: Employability Skills (60 Hours) .....                                                         | 38 |
| Assessment Guidelines and Weightage .....                                                                    | 45 |
| <i>Assessment Guidelines</i> .....                                                                           | 45 |
| <i>Assessment Weightage</i> .....                                                                            | 47 |
| Acronyms .....                                                                                               | 49 |
| Glossary .....                                                                                               | 52 |

## Qualification Pack

### MES/Q0521: AR/VR - Indiaskills Worldskills

#### Brief Job Description

The individual in this job will act as an architecture and overlays AR-VR solution to deliver experience of their project vision to prospective users for analysis, learning and game development purpose. It articulates 3D content products for Virtual & Augmented Reality application with smart technology driven system (hardware) which brings realistic views to users. The Technical Artist also acts as bridge between the artists and programmers, ensuring the art content and features are easily integrated into the environment.

#### Personal Attributes

The person must be able to work under the Technical Lead supervision and must have the ability to plan and write original theme according to the guidelines provided. The role requires excellent communication skills and collaborative abilities. The AR /VR Developer must be structured and result oriented with focus on quality and deliverables.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [MES/N0562: Prepare model and complete the texture as per real-time engines requirement- Worldskills](#)
2. [MES/N0564: Test 3D models in the real-time/game environment- Worldskills](#)
3. [MES/N0565: Artificial intelligence & machine learning- Worldskills](#)
4. [MES/N0567: Deploy Internet of things \(IoT\)- Worldskills](#)
5. [MES/N0568: Enterprise block chain- Worldskills](#)
6. [MES/N0569: Prepare computer generated models- Worldskills](#)
7. [MES/N0570: Add textures to models- Worldskills](#)
8. [MES/N0104: Maintain Workplace Health & Safety](#)
9. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

#### Qualification Pack (QP) Parameters

|        |                       |
|--------|-----------------------|
| Sector | Media & Entertainment |
|--------|-----------------------|

## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sub-Sector</b>                                         | Media and Entertainment                                                                                                                                                                                                                                                                 |
| <b>Occupation</b>                                         | Art and Design, Art and Design                                                                                                                                                                                                                                                          |
| <b>Country</b>                                            | India                                                                                                                                                                                                                                                                                   |
| <b>NSQF Level</b>                                         | 4.5                                                                                                                                                                                                                                                                                     |
| <b>Credits</b>                                            | 17                                                                                                                                                                                                                                                                                      |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/2166.0501                                                                                                                                                                                                                                                                      |
| <b>Minimum Educational Qualification &amp; Experience</b> | 2-year Diploma after 12th grade (in any field) (12th plus 2 Years Diploma in design) with NA of experience<br>OR<br>Pursuing 3rd year of UG (2nd Year of Graduation in related field) with NA of experience<br>OR<br>12th Class with 4 Years of experience 4 Years Experience in design |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                         |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                      |
| <b>Minimum Job Entry Age</b>                              | 16 Years                                                                                                                                                                                                                                                                                |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                      |
| <b>Next Review Date</b>                                   | 08/02/2026                                                                                                                                                                                                                                                                              |
| <b>NSQC Approval Date</b>                                 | 08/02/2024                                                                                                                                                                                                                                                                              |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                     |
| <b>Reference code on NQR</b>                              | QG-4.5-ME-01841-2024-V1-MESC                                                                                                                                                                                                                                                            |
| <b>NQR Version</b>                                        | 1.0                                                                                                                                                                                                                                                                                     |

## Qualification Pack

### MES/N0562: Prepare model and complete the texture as per real-time engines requirement- Worldskills

#### Description

Preparing a model and completing its texture for real-time engines involves optimizing the 3D model's geometry and UV mapping for efficiency and creating textures that meet the technical specifications of real-time rendering engines. This process ensures the model performs well in real-time applications such as games, simulations, or interactive experiences. It includes tasks like optimizing polygon count, organizing UV layouts efficiently, and creating textures that adhere to the engine's texture size and format requirements while maintaining visual quality and fidelity.

#### Scope

The scope covers the following :

- The scope of preparing a model and completing its texture for real-time engines involves optimizing the model's geometry and UV mapping for efficiency and creating textures that meet the technical specifications of the engine. This includes tasks like reducing polygon count, organizing UV layouts efficiently, and adhering to texture size and format requirements while maintaining visual quality.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

**PC1.** Analyse AR VR environments

**PC2.** Scribe design specifications and enterprise system architecture

**PC3.** Prepare digital models and asset content as per design specification

**PC4.** Use appropriate texturing and rendering systems to make displays as per design specification

**PC5.** Create character design/frame and structure the character as model

**PC6.** Create the visual style of the project, including colour sets, mood, etc

**PC7.** add textures to models to create photo-realistic models

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                          | 3            | 10              | -             | 2          |
| <b>PC1.</b> Analyse AR VR environments                                                                   | 1            | 1               | -             | -          |
| <b>PC2.</b> Scribe design specifications and enterprise system architecture                              | -            | 2               | -             | -          |
| <b>PC3.</b> Prepare digital models and asset content as per design specification                         | -            | 2               | -             | 1          |
| <b>PC4.</b> Use appropriate texturing and rendering systems to make displays as per design specification | 1            | 1               | -             | -          |
| <b>PC5.</b> Create character design/frame and structure the character as model                           | -            | 2               | -             | -          |
| <b>PC6.</b> Create the visual style of the project, including colour sets, mood, etc                     | 1            | 1               | -             | -          |
| <b>PC7.</b> add textures to models to create photo-realistic models                                      | -            | 1               | -             | 1          |
| <b>NOS Total</b>                                                                                         | <b>3</b>     | <b>10</b>       | <b>-</b>      | <b>2</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | MES/N0562                                                                                |
| <b>NOS Name</b>            | Prepare model and complete the texture as per real-time engines requirement- Worldskills |
| <b>Sector</b>              | Media & Entertainment                                                                    |
| <b>Sub-Sector</b>          |                                                                                          |
| <b>Occupation</b>          | Art and Design, Art and Design                                                           |
| <b>NSQF Level</b>          | 4.5                                                                                      |
| <b>Credits</b>             | 2                                                                                        |
| <b>Version</b>             | 1.0                                                                                      |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                                               |
| <b>Next Review Date</b>    | 08/02/2026                                                                               |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                                               |

## Qualification Pack

### MES/N0564: Test 3D models in the real-time/game environment- Worldskills

#### Description

Testing 3D models in real-time or game environments involves importing the models into the chosen engine or platform and evaluating their performance, functionality, and visual fidelity within the interactive environment. This process includes assessing factors such as model scale, animation integrity, texture rendering, lighting effects, collision detection, and overall compatibility with the target platform. It aims to ensure that the 3D models function as intended, providing an immersive and engaging experience for users within the real-time or game environment.

#### Scope

The scope covers the following :

- The scope of testing 3D models in real-time or game environments involves evaluating their performance, functionality, and visual fidelity within the context of the intended platform or engine. This includes assessing factors like model rendering, animation integrity, collision detection, texture quality, and overall scene optimization. Testing ensures that 3D models meet technical requirements, perform smoothly, and contribute positively to the immersive experience of the real-time or game environment.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Use AR VR Development tools
- PC2.** Understanding platform boundaries and optimisation techniques
- PC3.** Integrate models in the AR VR Project
- PC4.** Testing the models to ensure that they function correctly
- PC5.** Test the models to ensure that they meet the design specifications
- PC6.** production requirements and function as required
- PC7.** Check the continuity of models, textures or paintings and make sure they are fit for purpose of for all required camera positions and angles
- PC8.** Evaluate the quality of the assets in relation to others within the same context in which they will be used
- PC9.** Correct any problems or issues that may arise
- PC10.** Respond positively to feedback about work and changing textures/ other requirements and make refinements as needed



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                         | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                          | 3            | 10              | -             | 2          |
| <b>PC1.</b> Use AR VR Development tools                                                                                                                  | 1            | 1               | -             | -          |
| <b>PC2.</b> Understanding platform boundaries and optimisation techniques                                                                                | -            | 1               | -             | 1          |
| <b>PC3.</b> Integrate models in the AR VR Project                                                                                                        | -            | 1               | -             | -          |
| <b>PC4.</b> Testing the models to ensure that they function correctly                                                                                    | -            | 1               | -             | -          |
| <b>PC5.</b> Test the models to ensure that they meet the design specifications                                                                           | 1            | 1               | -             | -          |
| <b>PC6.</b> production requirements and function as required                                                                                             | -            | 1               | -             | 1          |
| <b>PC7.</b> Check the continuity of models, textures or paintings and make sure they are fit for purpose of for all required camera positions and angles | -            | 1               | -             | -          |
| <b>PC8.</b> Evaluate the quality of the assets in relation to others within the same context in which they will be used                                  | 1            | 1               | -             | -          |
| <b>PC9.</b> Correct any problems or issues that may arise                                                                                                | -            | 1               | -             | -          |
| <b>PC10.</b> Respond positively to feedback about work and changing textures/ other requirements and make refinements as needed                          | -            | 1               | -             | -          |
| <b>NOS Total</b>                                                                                                                                         | <b>3</b>     | <b>10</b>       | <b>-</b>      | <b>2</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                               |
|----------------------------|---------------------------------------------------------------|
| <b>NOS Code</b>            | MES/N0564                                                     |
| <b>NOS Name</b>            | Test 3D models in the real-time/game environment- Worldskills |
| <b>Sector</b>              | Media & Entertainment                                         |
| <b>Sub-Sector</b>          |                                                               |
| <b>Occupation</b>          | Art and Design, Art and Design                                |
| <b>NSQF Level</b>          | 4.5                                                           |
| <b>Credits</b>             | 2                                                             |
| <b>Version</b>             | 1.0                                                           |
| <b>Last Reviewed Date</b>  | 08/02/2024                                                    |
| <b>Next Review Date</b>    | 08/02/2026                                                    |
| <b>NSQC Clearance Date</b> | 08/02/2024                                                    |

## Qualification Pack

### MES/N0565: Artificial intelligence & machine learning- Worldskills

#### Description

Artificial intelligence (AI) and machine learning (ML) involve the development of computer systems that can perform tasks typically requiring human intelligence. AI focuses on creating systems that can mimic human intelligence, while ML is a subset of AI that focuses on algorithms and statistical models that enable computers to learn and improve from experience without being explicitly programmed. These technologies are applied across various domains, including natural language processing, computer vision, robotics, healthcare, finance, and more, revolutionizing industries and impacting everyday life.

#### Scope

The scope covers the following :

- Artificial intelligence (AI) and machine learning (ML) encompass the development of algorithms and systems that enable computers to perform tasks that typically require human intelligence. This includes understanding natural language, recognizing patterns, making predictions, and learning from data. The scope of AI and ML ranges from fundamental research in algorithms and data science to the application of these technologies in various fields such as healthcare, finance, transportation, and more. AI and ML are transforming industries by automating processes, optimizing decision-making, and unlocking new capabilities, ultimately shaping the future of technology and society.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify application of python in work environment and its process
- PC2.** Editing python using IDE (Integrated development environments) and text editors
- PC3.** Create python list
- PC4.** Apply python in developing 2D or 3D games, web & internet development, database access, network programming and business application.
- PC5.** Create Numpy and use large collection of high-level mathematical functions
- PC6.** Apply matplotlib
- PC7.** Apply mathematical rule: equations, functions and graphs; differentiation and optimization, vector and matrices, statistics and probability
- PC8.** Explore data for machine learning
- PC9.** Prepare data and clean data
- PC10.** Foundation of Artificial Intelligence (AI) in machine learning
- PC11.** Recognise languages – computer vision
- PC12.** Convert Bots as platform
- PC13.** Detect digits in hand-written digit image
- PC14.** Build a model to forecast a time data using a recurrent network
- PC15.** Build text data application using recurrent LSTM (long short term memory)
- PC16.** Create neural models for machine translation and conversion

## Qualification Pack

- PC17.** Create multimodal intelligence using languages
- PC18.** Recognise speech using basic signal processing
- PC19.** Create acoustic model and labelling
- PC20.** Decoding acoustic features into speech
- PC21.** Apply ethical and legal framework for the data profession
- PC22.** Approach to data and analytics problems including big data, data science and AI
- PC23.** Apply data methods for ethical and legal work in Analytics and AI
- PC24.** Apply dynamic programming
- PC25.** Explore, manipulate and analyse images using python packages for computer vision
- PC26.** Implement image classification using classical machine learning and deep learning techniques.
- PC27.** Use data augmentation and transfer learning to create highly-effective convolutional neural networks (CNNs)
- PC28.** Classify image to use object detection and semantic segmentation models. approach to data and analytics problems including big data, data science and AI

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                         | 3            | 5               | -             | 2          |
| <b>PC1.</b> Identify application of pythons in work environment and its process                                                                         | 1            | -               | -             | -          |
| <b>PC2.</b> Editing python using IDE (Integrated development environments) and text editors                                                             | -            | 1               | -             | -          |
| <b>PC3.</b> Create python list                                                                                                                          | -            | -               | -             | -          |
| <b>PC4.</b> Apply python in developing 2D or 3D games, web & internet development, database access, network programming and business application.       | -            | -               | -             | 1          |
| <b>PC5.</b> Create Numpy and use large collection of high-level mathematical functions                                                                  | -            | -               | -             | -          |
| <b>PC6.</b> Apply matplotlib                                                                                                                            | -            | -               | -             | -          |
| <b>PC7.</b> Apply mathematical rule: equations, functions and graphs; differentiation and optimization, vector and matrices, statistics and probability | -            | 1               | -             | -          |
| <b>PC8.</b> Explore data for machine learning                                                                                                           | 1            | -               | -             | -          |
| <b>PC9.</b> Prepare data and clean data                                                                                                                 | -            | -               | -             | -          |
| <b>PC10.</b> Foundation of Artificial Intelligence (AI) in machine learning                                                                             | -            | -               | -             | -          |
| <b>PC11.</b> Recognise languages – computer vision                                                                                                      | -            | -               | -             | -          |
| <b>PC12.</b> Convert Bots as platform                                                                                                                   | -            | 1               | -             | -          |
| <b>PC13.</b> Detect digits in hand-written digit image                                                                                                  | -            | -               | -             | -          |
| <b>PC14.</b> Build a model to forecast a time data using a recurrent network                                                                            | -            | -               | -             | -          |
| <b>PC15.</b> Build text data application using recurrent LSTM (long short term memory)                                                                  | -            | -               | -             | -          |
| <b>PC16.</b> Create neural models for machine translation and conversion                                                                                | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC17.</b> Create multimodel intelligence using languages                                                                                                           | -            | -               | -             | -          |
| <b>PC18.</b> Recognise speech using basic signal processing                                                                                                           | -            | -               | -             | -          |
| <b>PC19.</b> Create acoustic model and labelling                                                                                                                      | -            | -               | -             | -          |
| <b>PC20.</b> Decoding acoustic features into speech                                                                                                                   | -            | -               | -             | -          |
| <b>PC21.</b> Apply ethical and legal framework for the data profession                                                                                                | -            | 1               | -             | -          |
| <b>PC22.</b> Approach to data and analytics problems including big data, data science and AI                                                                          | -            | -               | -             | -          |
| <b>PC23.</b> Apply data methods for ethical and legal work in Analytics and AI                                                                                        | 1            | -               | -             | -          |
| <b>PC24.</b> Apply dynamic programming                                                                                                                                | -            | -               | -             | -          |
| <b>PC25.</b> Explore, manipulate and analyse images using python packages for computer vision                                                                         | -            | -               | -             | 1          |
| <b>PC26.</b> Implement image classification using classical machine learning and deep learning techniques.                                                            | -            | -               | -             | -          |
| <b>PC27.</b> Use data augmentation and transfer learning to create highly-effective convolutional neural networks (CNNs)                                              | -            | 1               | -             | -          |
| <b>PC28.</b> Classify image to use object detection and semantic segmentation models. approach to data and analytics problems including big data, data science and AI | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                      | <b>3</b>     | <b>5</b>        | <b>-</b>      | <b>2</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                         |
|----------------------------|---------------------------------------------------------|
| <b>NOS Code</b>            | MES/N0565                                               |
| <b>NOS Name</b>            | Artificial intelligence & machine learning- Worldskills |
| <b>Sector</b>              | Media & Entertainment                                   |
| <b>Sub-Sector</b>          |                                                         |
| <b>Occupation</b>          | Art and Design, Art and Design                          |
| <b>NSQF Level</b>          | 4.5                                                     |
| <b>Credits</b>             | 2                                                       |
| <b>Version</b>             | 1.0                                                     |
| <b>Last Reviewed Date</b>  | 08/02/2024                                              |
| <b>Next Review Date</b>    | 08/02/2026                                              |
| <b>NSQC Clearance Date</b> | 08/02/2024                                              |

## Qualification Pack

### MES/N0567: Deploy Internet of things (IoT)- Worldskills

#### Description

Deploying Internet of Things (IoT) involves implementing connected devices and systems to collect, transmit, and analyze data for various applications. It includes hardware setup, network configuration, software development, and data management. IoT deployment aims to improve efficiency, enable automation, and provide valuable insights across industries such as healthcare, manufacturing, agriculture, and smart cities.

#### Scope

The scope covers the following :

- The scope of deploying Internet of Things (IoT) involves implementing and integrating connected devices, sensors, and systems to gather, transmit, and analyze data in various environments. This encompasses hardware selection, network infrastructure setup, software development, and data management. IoT deployment aims to improve efficiency, enhance automation, enable real-time monitoring and control, and generate insights for informed decision-making across industries such as healthcare, manufacturing, agriculture, smart cities, and more.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** interpret IoT solution for dummies
- PC2.** Apply principles to follow for a successful deployment
- PC3.** Analyse IoT connectivity and related technologies
- PC4.** Select a board for prototyping
- PC5.** Analyse digital signage solutions for windows IoT platform
- PC6.** Run environment locally
- PC7.** Apply IoT Central, maps and an IoT SaaS solution
- PC8.** Customize UX and redeploy a microservice
- PC9.** Integrate with visualization tools
- PC10.** Use IoT Hub and connect MX Chip
- PC11.** Connect a Pi simulator to IoT Hub
- PC12.** Visualize time-series data with Time Series Insights
- PC13.** React to critical device lifecycle events and trigger Actions
- PC14.** Cold path storage and hot path analytics
- PC15.** Load test using Device Simulator and configure and monitor IoT devices at scale
- PC16.** Customize the Remote Monitoring solution accelerator
- PC17.** Host IoT solution accelerator
- PC18.** Scale IoT solution, IoT data and extract insights
- PC19.** Explore Edge intelligence in a Connected Factory



## Qualification Pack

- PC20.** Sequence IoT Hub primitives and Hub messaging
- PC21.** Host device management with IoT Hub and primitives
- PC22.** Examine automatic device management
- PC23.** Use IoT SDKs and developer tools
- PC24.** Analyse Hub device provisioning service

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                              | 3            | 5               | -             | 2          |
| <b>PC1.</b> interpret IoT solution for dummies                                               | 1            | -               | -             | -          |
| <b>PC2.</b> Apply principles to follow for a successful deployment                           | -            | 1               | -             | -          |
| <b>PC3.</b> Analyse IoT connectivity and related technologies                                | -            | -               | -             | -          |
| <b>PC4.</b> Select a board for prototyping                                                   | -            | -               | -             | -          |
| <b>PC5.</b> Analyse digital signage solutions for windows IoT platform                       | -            | -               | -             | -          |
| <b>PC6.</b> Run environment locally                                                          | -            | -               | -             | -          |
| <b>PC7.</b> Apply IoT Central, maps and an IoT SaaS solution                                 | -            | -               | -             | -          |
| <b>PC8.</b> Customize UX and redeploy a microservice                                         | -            | -               | -             | 1          |
| <b>PC9.</b> Integrate with visualization tools                                               | -            | -               | -             | -          |
| <b>PC10.</b> Use IoT Hub and connect MX Chip                                                 | 1            | -               | -             | -          |
| <b>PC11.</b> Connect a Pi simulator to IoT Hub                                               | -            | 1               | -             | -          |
| <b>PC12.</b> Visualize time-series data with Time Series Insights                            | -            | -               | -             | -          |
| <b>PC13.</b> React to critical device lifecycle events and trigger Actions                   | -            | -               | -             | -          |
| <b>PC14.</b> Cold path storage and hot path analytics                                        | -            | -               | -             | -          |
| <b>PC15.</b> Load test using Device Simulator and configure and monitor IoT devices at scale | -            | -               | -             | -          |
| <b>PC16.</b> Customize the Remote Monitoring solution accelerator                            | -            | 1               | -             | -          |
| <b>PC17.</b> Host IoT solution accelerator                                                   | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC18.</b> Scale IoT solution, IoT data and extract insights  | -            | -               | -             | 1          |
| <b>PC19.</b> Explore Edge intelligence in a Connected Factory   | -            | -               | -             | -          |
| <b>PC20.</b> Sequence IoT Hub primitives and Hub messaging      | -            | 1               | -             | -          |
| <b>PC21.</b> Host device management with IoT Hub and primitives | -            | -               | -             | -          |
| <b>PC22.</b> Examine automatic device management                | 1            | -               | -             | -          |
| <b>PC23.</b> Use IoT SDKs and developer tools                   | -            | -               | -             | -          |
| <b>PC24.</b> Analyse Hub device provisioning service            | -            | 1               | -             | -          |
| <b>NOS Total</b>                                                | <b>3</b>     | <b>5</b>        | <b>-</b>      | <b>2</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                              |
|----------------------------|----------------------------------------------|
| <b>NOS Code</b>            | MES/N0567                                    |
| <b>NOS Name</b>            | Deploy Internet of things (IoT)- Worldskills |
| <b>Sector</b>              | Media & Entertainment                        |
| <b>Sub-Sector</b>          |                                              |
| <b>Occupation</b>          | Art and Design, Art and Design               |
| <b>NSQF Level</b>          | 4.5                                          |
| <b>Credits</b>             | 2                                            |
| <b>Version</b>             | 1.0                                          |
| <b>Last Reviewed Date</b>  | 08/02/2024                                   |
| <b>Next Review Date</b>    | 08/02/2026                                   |
| <b>NSQC Clearance Date</b> | 08/02/2024                                   |

## Qualification Pack

### MES/N0568: Enterprise block chain- Worldskills

#### Description

Enterprise blockchain involves implementing distributed ledger technology within businesses or organizations to securely record, store, and manage transactions or data across a network of participants. Unlike public blockchains like Bitcoin or Ethereum, enterprise blockchains are typically permissioned, meaning that access to the network is controlled and restricted to authorized entities. This technology enables organizations to streamline processes, enhance transparency, reduce costs, mitigate fraud, and improve trust among stakeholders. It finds applications in various industries, including finance, supply chain management, healthcare, and government, among others.

#### Scope

The scope covers the following :

- The scope of enterprise blockchain involves implementing distributed ledger technology within organizations to streamline processes, enhance transparency, and improve trust and security in data transactions. It includes designing and deploying blockchain networks tailored to enterprise needs, developing smart contracts, integrating blockchain with existing systems, and ensuring compliance with regulations. Enterprise blockchain applications span industries such as supply chain management, finance, healthcare, and government, offering benefits such as increased efficiency, reduced costs, and enhanced data integrity and auditability.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret block chain concept
- PC2.** Cryptocurrencies and risks
- PC3.** Consensus algorithms
- PC4.** Transact block chain
- PC5.** Synergy of blockchain with other cutting edge technologies
- PC6.** Design block chain
- PC7.** Analyse ethereum
- PC8.** Design network structure of ethereum
- PC9.** Installation of toolchain
- PC10.** Setting up private node
- PC11.** Building blocks of blockchain solutions
- PC12.** Architect HLF runtime
- PC13.** Installation of Hyperledger fabric
- PC14.** Configure Hyperledger Fabric
- PC15.** Analyse the role of System components
- PC16.** Use-Case Introduction and create Hyperledger Fabric Blockchain network
- PC17.** Implement Smart Contract / Chain code

## Qualification Pack

- PC18.** Install and Instantiate chain code
- PC19.** Deploy Client Application (DApp)
- PC20.** Communicate Transport Layer Security (TLS)
- PC21.** Architect security, threat and mitigation

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                     | 3            | 5               | -             | 2          |
| <b>PC1.</b> Interpret block chain concept                                           | 1            | -               | -             | -          |
| <b>PC2.</b> Cryptocurrencies and risks                                              | -            | 1               | -             | -          |
| <b>PC3.</b> Consensus algorithms                                                    | -            | -               | -             | -          |
| <b>PC4.</b> Transact block chain                                                    | -            | -               | -             | -          |
| <b>PC5.</b> Synergy of blockchain with other cutting edge technologies              | -            | -               | -             | -          |
| <b>PC6.</b> Design block chain                                                      | -            | 1               | -             | -          |
| <b>PC7.</b> Analyse ethereum                                                        | -            | -               | -             | -          |
| <b>PC8.</b> Design network structure of ethereum                                    | 1            | -               | -             | -          |
| <b>PC9.</b> Installation of toolchain                                               | -            | -               | -             | -          |
| <b>PC10.</b> Setting up private node                                                | -            | -               | -             | -          |
| <b>PC11.</b> Building blocks of blockchain solutions                                | -            | -               | -             | -          |
| <b>PC12.</b> Architect HLF runtime                                                  | -            | -               | -             | -          |
| <b>PC13.</b> Installation of Hyperledger fabric                                     | -            | 1               | -             | -          |
| <b>PC14.</b> Configure Hyperledger Fabric                                           | -            | -               | -             | -          |
| <b>PC15.</b> Analyse the role of System components                                  | -            | -               | -             | -          |
| <b>PC16.</b> Use-Case Introduction and create Hyperledger Fabric Blockchain network | -            | 1               | -             | 1          |
| <b>PC17.</b> Implement Smart Contract / Chain code                                  | -            | -               | -             | -          |
| <b>PC18.</b> Install and Instantiate chain code                                     | 1            | -               | -             | -          |
| <b>PC19.</b> Deploy Client Application (DApp)                                       | -            | -               | -             | -          |
| <b>PC20.</b> Communicate Transport Layer Security (TLS)                             | -            | 1               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC21.</b> Architect security, threat and mitigation | -            | -               | -             | 1          |
| <b>NOS Total</b>                                       | <b>3</b>     | <b>5</b>        | <b>-</b>      | <b>2</b>   |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                     |
|----------------------------|-------------------------------------|
| <b>NOS Code</b>            | MES/N0568                           |
| <b>NOS Name</b>            | Enterprise block chain- Worldskills |
| <b>Sector</b>              | Media & Entertainment               |
| <b>Sub-Sector</b>          |                                     |
| <b>Occupation</b>          | Art and Design, Art and Design      |
| <b>NSQF Level</b>          | 4.5                                 |
| <b>Credits</b>             | 2                                   |
| <b>Version</b>             | 1.0                                 |
| <b>Last Reviewed Date</b>  | 08/02/2024                          |
| <b>Next Review Date</b>    | 08/02/2026                          |
| <b>NSQC Clearance Date</b> | 08/02/2024                          |

## Qualification Pack

### MES/N0569: Prepare computer generated models- Worldskills

#### Description

Preparing computer-generated models involves creating digital representations of objects, characters, environments, or scenes using specialized software. This process includes modeling, texturing, rigging (if applicable), and animating the models to achieve desired visual outcomes. Computer-generated models are used in various industries, including animation, gaming, architecture, product design, and visual effects, to visualize concepts, prototypes, or final products. They play a crucial role in creating immersive digital experiences and communicating ideas effectively.

#### Scope

The scope covers the following :

- The scope of preparing computer-generated models involves creating digital representations of objects, characters, environments, or scenes using specialized software. This encompasses tasks such as modeling, texturing, rigging, and animation to achieve desired visual effects and realism. Computer-generated models are used in various industries including film, gaming, architecture, product design, and simulation for visualization, prototyping, and production purposes. The process involves collaboration between artists, designers, and technical experts to bring creative visions to life in digital form.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare digital models according to the design brief (appearance, complexion, dressing, moods, personalities, expressions etc.), requirements (number, types, duplicates etc.) and specifications (dimensions, operating parameters etc.)
- PC2.** Create prototypes/pilots for testing
- PC3.** Understand the final display medium and adapt / suggest the model for its polycounts, mesh complexity, movement capability etc.
- PC4.** Ensure that the models will be able to perform properly once animated, are uniform and consistent and are delivered in appropriate formats that can be used by others

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                       | 3            | 10              | -             | 2          |
| <b>PC1.</b> Prepare digital models according to the design brief (appearance, complexion, dressing, moods, personalities, expressions etc.), requirements (number, types, duplicates etc.) and specifications (dimensions, operating parameters etc.) | 1            | 3               | -             | 1          |
| <b>PC2.</b> Create prototypes/pilots for testing                                                                                                                                                                                                      | 1            | 3               | -             | -          |
| <b>PC3.</b> Understand the final display medium and adapt / suggest the model for its polycounts, mesh complexity, movement capability etc.                                                                                                           | 1            | 2               | -             | -          |
| <b>PC4.</b> Ensure that the models will be able to perform properly once animated, are uniform and consistent and are delivered in appropriate formats that can be used by others                                                                     | -            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                                                                                                      | <b>3</b>     | <b>10</b>       | <b>-</b>      | <b>2</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                |
|----------------------------|------------------------------------------------|
| <b>NOS Code</b>            | MES/N0569                                      |
| <b>NOS Name</b>            | Prepare computer generated models- Worldskills |
| <b>Sector</b>              | Media & Entertainment                          |
| <b>Sub-Sector</b>          |                                                |
| <b>Occupation</b>          | Art and Design, Art and Design                 |
| <b>NSQF Level</b>          | 4.5                                            |
| <b>Credits</b>             | 2                                              |
| <b>Version</b>             | 1.0                                            |
| <b>Last Reviewed Date</b>  | 08/02/2024                                     |
| <b>Next Review Date</b>    | 08/02/2026                                     |
| <b>NSQF Clearance Date</b> | 08/02/2024                                     |

## Qualification Pack

### MES/N0570: Add textures to models- Worldskills

#### Description

Adding textures to models involves applying surface details, colors, and patterns to 3D objects to enhance their visual appearance and realism. This process enhances the overall aesthetics of the model, making it more visually appealing and immersive. Textures can simulate various materials such as wood, metal, fabric, or skin, adding depth and texture to the surfaces. Texturing is essential in various industries including gaming, animation, architecture, and product design, as it helps create lifelike and engaging virtual environments.

#### Scope

The scope covers the following :

- Adding textures to models involves applying surface details, colors, and patterns to 3D objects to enhance their visual appearance and realism. This process enhances the overall aesthetics of the model, making it more visually appealing and immersive. Textures can simulate various materials such as wood, metal, fabric, or skin, adding depth and texture to the surfaces. Texturing is essential in various industries including gaming, animation, architecture, and product design, as it helps create lifelike and engaging virtual environments.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Visualize possibilities for adding textures to models to create photorealistic models/images
- PC2.** Develop and add textures to models in accordance to the design brief and concept art for different types of models
- PC3.** Understand the final exhibition medium and adapt the textures accordingly
- PC4.** Manage quality of textures during the animation process and ensure uniformity and consistency in the final output
- PC5.** supply work in appropriate formats that can be used by others in the pipeline

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                | 3            | 10              | -             | 2          |
| <b>PC1.</b> Visualize possibilities for adding textures to models to create photorealistic models/images                       | 1            | 2               | -             | -          |
| <b>PC2.</b> Develop and add textures to models in accordance to the design brief and concept art for different types of models | -            | 2               | -             | 1          |
| <b>PC3.</b> Understand the final exhibition medium and adapt the textures accordingly                                          | 1            | 2               | -             | -          |
| <b>PC4.</b> Manage quality of textures during the animation process and ensure uniformity and consistency in the final output  | -            | 2               | -             | 1          |
| <b>PC5.</b> supply work in appropriate formats that can be used by others in the pipeline                                      | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                                               | <b>3</b>     | <b>10</b>       | <b>-</b>      | <b>2</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                     |
|----------------------------|-------------------------------------|
| <b>NOS Code</b>            | MES/N0570                           |
| <b>NOS Name</b>            | Add textures to models- Worldskills |
| <b>Sector</b>              | Media & Entertainment               |
| <b>Sub-Sector</b>          |                                     |
| <b>Occupation</b>          | Art and Design, Art and Design      |
| <b>NSQF Level</b>          | 4.5                                 |
| <b>Credits</b>             | 2                                   |
| <b>Version</b>             | 1.0                                 |
| <b>Last Reviewed Date</b>  | 08/02/2024                          |
| <b>Next Review Date</b>    | 08/02/2026                          |
| <b>NSQC Clearance Date</b> | 08/02/2024                          |

## Qualification Pack

### MES/N0104: Maintain Workplace Health & Safety

#### Description

This OS unit is about contributing towards maintaining a healthy, safe and secure working environment

#### Elements and Performance Criteria

##### *Understanding the health, safety and security risks prevalent in the workplace*

To be competent, the user/individual on the job must be able to:

- PC1.** understand and comply with the organizations current health, safety and security policies and procedures
- PC2.** understand the safe working practices pertaining to own occupation
- PC3.** understand the government norms and policies relating to health and safety including emergency procedures for illness, accidents, fires or others which may involve evacuation of the premises
- PC4.** participate in organization health and safety knowledge sessions and drills

##### *Knowing the people responsible for health and safety and the resources available*

To be competent, the user/individual on the job must be able to:

- PC5.** identify the people responsible for health and safety in the workplace, including those to contact in case of an emergency
- PC6.** identify security signals e.g. fire alarms and places such as staircases, fire warden stations, first aid and medical rooms

##### *Identifying and reporting risks*

To be competent, the user/individual on the job must be able to:

- PC7.** identify aspects of your workplace that could cause potential risk to own and others health and safety
- PC8.** ensure own personal health and safety, and that of others in the workplace through precautionary measures
- PC9.** identify and recommend opportunities for improving health, safety, and security to the designated person
- PC10.** report any hazards outside the individuals authority to the relevant person in line with organizational procedures and warn other people who may be affected

##### *Complying with procedures in the event of an emergency*

To be competent, the user/individual on the job must be able to:

- PC11.** follow organizations emergency procedures for accidents, fires or any other natural calamity in case of a hazard
- PC12.** identify and correct risks like illness, accidents, fires or any other natural calamity safely and within the limits of individuals authority

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



## Qualification Pack

- KU1.** Organizations norms and policies relating to health and safety
- KU2.** Government norms and policies regarding health and safety and related emergency procedures
- KU3.** Limits of authority while dealing with risks/ hazards
- KU4.** The importance of maintaining high standards of health and safety at a workplace
- KU5.** The different types of health and safety hazards in a workplace
- KU6.** Safe working practices for own job role
- KU7.** Evacuation procedures and other arrangements for handling risks
- KU8.** Names and contact numbers of people responsible for health and safety in a workplace
- KU9.** How to summon medical assistance and the emergency services, where necessary
- KU10.** Vendors or manufacturers instructions for maintaining health and safety while using equipment, systems and/or machines

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** how to write and provide feedback regarding health and safety to the concerned people
- GS2.** how to write and highlight potential risks or report a hazard to the concerned people
- GS3.** read instructions, policies, procedures and norms relating to health and safety
- GS4.** highlight potential risks and report hazards to the designated people
- GS5.** listen and communicate information with all anyone concerned or affected
- GS6.** make decisions on a suitable course of action or plan
- GS7.** plan and organize people and resources to deal with risks/ hazards that lie within the scope of ones individual authority
- GS8.** apply problem solving approaches in different situations
- GS9.** understand hazards that fall within the scope of individual authority and report all hazards that may supersede ones authority
- GS10.** apply balanced judgments in different situations
- GS11.** How to write and provide feedback regarding health and safety to the concerned people
- GS12.** How to write and highlight potential risks or report a hazard to the concerned people
- GS13.** Read instructions, policies, procedures and norms relating to health and safety
- GS14.** Highlight potential risks and report hazards to the designated people
- GS15.** Listen and communicate information with all anyone concerned or affected
- GS16.** Make decisions on a suitable course of action or plan
- GS17.** Plan and organize people and resources to deal with risks/ hazards that lie within the scope of ones individual authority
- GS18.** Apply problem solving approaches in different situations
- GS19.** build and maintain positive and effective relationships with colleges and customers
- GS20.** analyze data and activities
- GS21.** Understand hazards that fall within the scope of individual authority and report all hazards that may supersede ones authority



## Qualification Pack

**GS22.** Apply balanced judgments in different situations

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Understanding the health, safety and security risks prevalent in the workplace</i>                                                                                                                      | <b>15</b>    | <b>15</b>       | -             | -          |
| <b>PC1.</b> understand and comply with the organizations current health, safety and security policies and procedures                                                                                       | 5            | 5               | -             | -          |
| <b>PC2.</b> understand the safe working practices pertaining to own occupation                                                                                                                             | 5            | 5               | -             | -          |
| <b>PC3.</b> understand the government norms and policies relating to health and safety including emergency procedures for illness, accidents, fires or others which may involve evacuation of the premises | 3            | 2               | -             | -          |
| <b>PC4.</b> participate in organization health and safety knowledge sessions and drills                                                                                                                    | 2            | 3               | -             | -          |
| <i>Knowing the people responsible for health and safety and the resources available</i>                                                                                                                    | <b>10</b>    | <b>10</b>       | -             | -          |
| <b>PC5.</b> identify the people responsible for health and safety in the workplace, including those to contact in case of an emergency                                                                     | 5            | 5               | -             | -          |
| <b>PC6.</b> identify security signals e.g. fire alarms and places such as staircases, fire warden stations, first aid and medical rooms                                                                    | 5            | 5               | -             | -          |
| <i>Identifying and reporting risks</i>                                                                                                                                                                     | <b>18</b>    | <b>17</b>       | -             | -          |
| <b>PC7.</b> identify aspects of your workplace that could cause potential risk to own and others health and safety                                                                                         | 5            | 5               | -             | -          |
| <b>PC8.</b> ensure own personal health and safety, and that of others in the workplace through precautionary measures                                                                                      | 5            | 5               | -             | -          |
| <b>PC9.</b> identify and recommend opportunities for improving health, safety, and security to the designated person                                                                                       | 3            | 2               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> report any hazards outside the individuals authority to the relevant person in line with organizational procedures and warn other people who may be affected | 5            | 5               | -             | -          |
| <i>Complying with procedures in the event of an emergency</i>                                                                                                             | <b>7</b>     | <b>8</b>        | -             | -          |
| <b>PC11.</b> follow organizations emergency procedures for accidents, fires or any other natural calamity in case of a hazard                                             | 5            | 5               | -             | -          |
| <b>PC12.</b> identify and correct risks like illness, accidents, fires or any other natural calamity safely and within the limits of individuals authority                | 2            | 3               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                          | <b>50</b>    | <b>50</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                         |
|----------------------------|---------------------------------------------------------|
| <b>NOS Code</b>            | MES/N0104                                               |
| <b>NOS Name</b>            | Maintain Workplace Health & Safety                      |
| <b>Sector</b>              | Media & Entertainment                                   |
| <b>Sub-Sector</b>          | Film, Television, Animation, Gaming, Radio, Advertising |
| <b>Occupation</b>          | Ad sales/Account Management/Scheduling/Traffic          |
| <b>NSQF Level</b>          | 5                                                       |
| <b>Credits</b>             | 2                                                       |
| <b>Version</b>             | 1.0                                                     |
| <b>Last Reviewed Date</b>  | 10/09/2022                                              |
| <b>Next Review Date</b>    | 29/06/2027                                              |
| <b>NSQC Clearance Date</b> | 10/09/2022                                              |

## Qualification Pack

### DGT/VSQ/N0102: Employability Skills (60 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

## Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

## Qualification Pack

**PC28.** follow appropriate hygiene and grooming standards

*Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC29.** create a professional Curriculum vitae (Résumé)

**PC30.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

**PC31.** apply to identified job openings using offline /online methods as per requirement

**PC32.** answer questions politely, with clarity and confidence, during recruitment and selection

**PC33.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills and different learning and employability related portals

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

**KU6.** importance of career development and setting long- and short-term goals

**KU7.** about effective communication

**KU8.** POSH Act

**KU9.** Gender sensitivity and inclusivity

**KU10.** different types of financial institutes, products, and services

**KU11.** how to compute income and expenditure

**KU12.** importance of maintaining safety and security in offline and online financial transactions

**KU13.** different legal rights and laws

**KU14.** different types of digital devices and the procedure to operate them safely and securely

**KU15.** how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

**KU16.** how to identify business opportunities

**KU17.** types and needs of customers

**KU18.** how to apply for a job and prepare for an interview

**KU19.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

**GS1.** read and write different types of documents/instructions/correspondence

**GS2.** communicate effectively using appropriate language in formal and informal settings



## Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> identify employability skills required for jobs in various industries                                                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability portals                                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC3.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC4.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>2</b>     | <b>4</b>        | -             | -          |
| <b>PC5.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC6.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC7.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |
| <b>PC8.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC9.</b> write short messages, notes, letters, e-mails etc. in English                                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                                                                                                                                                                     | <b>1</b>     | <b>2</b>        | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> understand the difference between job and career                                                         | -            | -               | -             | -          |
| <b>PC11.</b> prepare a career development plan with short- and long-term goals, based on aptitude                     | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> follow verbal and non-verbal communication etiquette and active listening techniques in various settings | -            | -               | -             | -          |
| <b>PC13.</b> work collaboratively with others in a team                                                               | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                      | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> communicate and behave appropriately with all genders and PwD                                            | -            | -               | -             | -          |
| <b>PC15.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                      | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                   | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC16.</b> select financial institutions, products and services as per requirement                                  | -            | -               | -             | -          |
| <b>PC17.</b> carry out offline and online financial transactions, safely and securely                                 | -            | -               | -             | -          |
| <b>PC18.</b> identify common components of salary and compute income, expenses, taxes, investments etc                | -            | -               | -             | -          |
| <b>PC19.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                 | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                       | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC20.</b> operate digital devices and carry out basic internet operations securely and safely                      | -            | -               | -             | -          |
| <b>PC21.</b> use e- mail and social media platforms and virtual collaboration tools to work effectively               | -            | -               | -             | -          |
| <b>PC22.</b> use basic features of word processor, spreadsheets, and presentations                                    | -            | -               | -             | -          |

## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Entrepreneurship</i>                                                                                                                                                          | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC23.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                       | -            | -               | -             | -          |
| <b>PC24.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                      | -            | -               | -             | -          |
| <b>PC25.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                           | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                          | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC26.</b> identify different types of customers                                                                                                                               | -            | -               | -             | -          |
| <b>PC27.</b> identify and respond to customer requests and needs in a professional manner.                                                                                       | -            | -               | -             | -          |
| <b>PC28.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC29.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC30.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC31.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC32.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC33.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0102                   |
| <b>NOS Name</b>            | Employability Skills (60 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 4                               |
| <b>Credits</b>             | 2                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 18/02/2025                      |
| <b>Next Review Date</b>    | 18/02/2028                      |
| <b>NSQC Clearance Date</b> | 18/02/2025                      |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

#### 1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SDMS/SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping SSC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- SSC monitors the assessment process & records

#### 2. Testing Environment:

- Confirm that the centre is available at the same address as mentioned on SDMS or SIP
- Check the duration of the training.
- Check the Assessment Start and End time to be as 10 a.m. and 5 p.m.

## Qualification Pack

- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.
- Check the mode of assessment—Online (TAB/Computer) or Offline (OMR/PP).
- Confirm the number of TABs on the ground are correct to execute the Assessment smoothly.
- Check the availability of the Lab Equipment for the particular Job Role.

### 3. Assessment Quality Assurance levels / Framework:

- Question papers created by the Subject Matter Experts (SME)
- Question papers created by the SME verified by the other subject Matter Experts
- Questions are mapped with NOS and PC
- Question papers are prepared considering that level 1 to 3 is for the unskilled & semi-skilled individuals, and level 4 and above are for the skilled, supervisor & higher management.
- Assessor must be ToA certified & trainer must be ToT Certified
- Assessment agency must follow the assessment guidelines to conduct the assessment.

### 4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding
- Biometric or manual attendance sheet (stamped by TP) of the trainees during the training period
- Time-stamped & geotagged assessment (Theory + Viva + Practical) photographs & videos

### 5. Method of verification or validation:

- Surprise visit to the assessment location
- Random audit of the batch
- Random audit of any candidate

### 6. Method for assessment documentation, archiving, and access

## Qualification Pack

- Hard copies of the documents are stored
- Soft copies of the documents & photographs of the assessment are uploaded / accessed from Cloud Storage
- Soft copies of the documents & photographs of the assessment are stored in the Hard Drives

### Minimum Aggregate Passing % at QP Level : 70

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

### Compulsory NOS

| National Occupational Standards                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage |
|----------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|-----------|
| MES/N0562.Prepare model and complete the texture as per real-time engines requirement- Worldskills | 3            | 10              | 0             | 2          | 15          | 15        |
| MES/N0564.Test 3D models in the real-time/game environment- Worldskills                            | 3            | 10              | 0             | 2          | 15          | 10        |
| MES/N0565.Artificial intelligence & machine learning- Worldskills                                  | 3            | 5               | 0             | 2          | 10          | 15        |
| MES/N0567.Deploy Internet of things (IoT)- Worldskills                                             | 3            | 5               | 0             | 2          | 10          | 10        |
| MES/N0568.Enterprise block chain- Worldskills                                                      | 3            | 5               | 0             | 2          | 10          | 10        |
| MES/N0569.Prepare computer generated models- Worldskills                                           | 3            | 10              | 0             | 2          | 15          | 10        |
| MES/N0570.Add textures to models- Worldskills                                                      | 3            | 10              | 0             | 2          | 15          | 10        |
| MES/N0104.Maintain Workplace Health & Safety                                                       | 50           | 50              | -             | -          | 100         | 10        |

### Qualification Pack

| National Occupational Standards               | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|-----------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| DGT/VSQ/N0102.Employability Skills (60 Hours) | 20           | 30              | -             | -          | 50          | 10         |
| <b>Total</b>                                  | <b>91</b>    | <b>135</b>      | <b>-</b>      | <b>14</b>  | <b>240</b>  | <b>100</b> |



## Qualification Pack

### Acronyms

|             |                                                      |
|-------------|------------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)                    |
| <b>NSQF</b> | National Skills Qualifications Framework             |
| <b>QP</b>   | Qualifications Pack                                  |
| <b>TVET</b> | Technical and Vocational Education and Training      |
| <b>AA</b>   | Assessment Agency                                    |
| <b>AB</b>   | Awarding Body                                        |
| <b>ISCO</b> | International Standard Classification of Occupations |
| <b>NCO</b>  | National Classification of Occupations               |
| <b>NCrF</b> | National Credit Framework                            |
| <b>NOS</b>  | National Occupational Standard(s)                    |
| <b>NQR</b>  | National Qualification Register                      |
| <b>NSQF</b> | National Skills Qualifications Framework             |
| <b>OJT</b>  | On the Job Training                                  |
| <b>AA</b>   | Assessment Agency                                    |
| <b>AB</b>   | Awarding Body                                        |
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| <b>NCO</b>  | National Classification of Occupations               |
| <b>NCrF</b> | National Credit Framework                            |
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| <b>NSQF</b> | National Skills Qualifications Framework             |
| <b>OJT</b>  | On the Job Training                                  |
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## Qualification Pack

|             |                                                      |
|-------------|------------------------------------------------------|
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| <b>NQR</b>  | National Qualification Register                      |
| <b>NSQF</b> | National Skills Qualifications Framework             |
| <b>OJT</b>  | On the Job Training                                  |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |
| <b>Knowledge and Understanding (KU)</b>      | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                         |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |
| <b>National Occupational Standard</b>   | NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.                                                                                                                                                                               |
| <b>Qualification</b>                    | A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards                                                                                                                                                                                       |
| <b>Qualification File</b>               | A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.                                                                                                                                 |
| <b>Sector</b>                           | A grouping of professional activities on the basis of their main economic function, product, service or technology.                                                                                                                                                                                                                                                    |
| <b>Long Term Training</b>               | Long-term skilling means any vocational training program undertaken for a year and above. <a href="https://ncvet.gov.in/sites/default/files/NCVET.pdf">https://ncvet.gov.in/sites/default/files/NCVET.pdf</a>                                                                                                                                                          |
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## Qualification Pack

|                                       |                                                                                                                                                                                                                                        |
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## Qualification Pack

|                                       |                                                                                                                                                                                                                                                           |
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